



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 17/01/2020*

PROBLEM

We observe N samples of a realization of the random process $X[n] = S[n] + W[n]$, for $n=0, 1, \dots, N-1$. where $S[n] = A$ is the constant signal of interest (SOI), modelled as a random parameter with mean η_A and variance σ_A^2 , and $W[n]$ is zero mean additive noise, modelled as an Autoregressive process of order 1 with one-lag correlation coefficient ρ and variance σ_w^2 . The signal to noise ratio (SNR) is defined as follows:

$$SNR \triangleq \frac{E\{A^2\}}{E\{W^2[n]\}} = \frac{\eta_A^2 + \sigma_A^2}{\sigma_w^2} = \gamma \left(1 + \frac{\eta_A^2}{\sigma_A^2} \right), \text{ where } \gamma \triangleq \frac{\sigma_A^2}{\sigma_w^2}.$$

- (1) Derive the autocorrelation function (ACF) $R_X[m] = E\{X[n]X[n+m]\}$ and the power spectral density (PSD) $S_X(e^{j2\pi f}) = FT\{R_X[m]\}$ of the observed signal $X[n]$ and plot them.
- (2) Derive the optimal linear estimator, i.e. LMMSE estimator, of $S[n+1]$ given the observation of $X[n]$ and $X[n-1]$ (i.e. we observe $N=2$ samples) and express it as a function of γ and ρ . Show that by defining properly the parameter α , the estimator can be put in the form:

$$\hat{S}_{LMMSE}[n+1] = \alpha \bar{X} + (1-\alpha)\eta_A, \text{ where } \bar{X} \triangleq \frac{X[n] + X[n-1]}{2}.$$

- (3) Derive the mean square error (MSE) of $\hat{S}_{LMMSE}[n+1]$ and find the minimum value of γ which guarantees an MSE lower than 10^{-3} when $\sigma_A^2=1$ and $\rho=0$.
- (4) Assuming $\sigma_A^2=1$, plot the MSE as a function of γ (for $\rho=0$) and as a function of ρ (for $\gamma=1$) and finally analyze and comment the behavior of the estimation algorithm and of the MSE when σ_w^2 tends to 0 or $+\infty$ and ρ tends to -1, 0, or 1.



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Solution

(1) We can prove easily that the ACF of the process $S[n]=A$ is given by $R_S[m] = E\{A^2\} = \eta_A^2 + \sigma_A^2$.

The ACF of $W[n]$ is instead given by $R_W[m] = \sigma_W^2 \rho^{|m|}$, so we have:

$$R_X[m] = R_S[m] + R_W[m] = \eta_A^2 + \sigma_A^2 + \sigma_W^2 \rho^{|m|}.$$

The PSD is the Fourier Transform (FT) of the ACF:

$$\begin{aligned} S_X(e^{j2\pi f}) &= S_S(e^{j2\pi f}) + S_W(e^{j2\pi f}) \\ &= 2\pi(\eta_A^2 + \sigma_A^2)\delta(e^{j2\pi f} - 1) + \sigma_W^2(1 - \rho^2) \left| \frac{1}{1 - \rho e^{-j2\pi f}} \right|^2 \\ &= 2\pi(\eta_A^2 + \sigma_A^2)\delta(e^{j2\pi f} - 1) + \frac{\sigma_W^2(1 - \rho^2)}{1 + \rho^2 - 2\rho \cos(2\pi f)} \end{aligned}$$

(2) The LMMSE estimator of $S[n+1]=A$ given the observed data $X[n]$ and $X[n-1]$ is the FIR(2) Wiener filter that can be derived by imposing that the estimator is unbiased and by applying the Orthogonality principle:

$$\hat{S}_{LMMSE}[n+1] = \hat{A}_{LMMSE} = \mathbf{a}^T \mathbf{X} + b = a_1 X[n] + a_2 X[n-1] + b, \text{ where } \mathbf{X} = \begin{bmatrix} X[n] \\ X[n-1] \end{bmatrix}$$

$$\varepsilon[n+1] = S[n+1] - \hat{S}_{LMMSE}[n+1] = A - \hat{A}_{LMMSE} = A - a_1 X[n] - a_2 X[n-1] - b$$

$$E\{\varepsilon[n+1]\} = 0 \Rightarrow b = (1 - a_1 - a_2)\eta_A$$

$$E\{\varepsilon[n+1]X[n-m]\} = E\{(S[n+1] - a_1 X[n] - a_2 X[n-1] - b)X[n-m]\} = 0, \quad m = 0, 1$$

$$= E\{(S[n+1] - \eta_A - a_1(X[n] - \eta_A) - a_2(X[n-1] - \eta_A))X[n-m]\}$$

$$C_{SX}[-(m+1)] = a_1 C_X[m] + a_2 C_X[m-1], \quad m = 0, 1$$

$$C_{SX}[-(m+1)] = C_S[m+1] = R_S[m+1] - \eta_A^2 = (\eta_A^2 + \sigma_A^2) - \eta_A^2 = \sigma_A^2$$

$$C_X[m] = R_X[m] - \eta_A^2 = \sigma_A^2 + \sigma_W^2 \rho^{|m|}$$



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$$\begin{bmatrix} C_X[0] & C_X[-1] \\ C_X[1] & C_X[0] \end{bmatrix} \mathbf{a} = \begin{bmatrix} C_{SX}[-1] \\ C_{SX}[-2] \end{bmatrix} = \sigma_A^2 \mathbf{1}$$

$$\mathbf{a} = \begin{bmatrix} C_X[0] & C_X[-1] \\ C_X[1] & C_X[0] \end{bmatrix}^{-1} \sigma_A^2 \mathbf{1} = \begin{bmatrix} \sigma_A^2 + \sigma_W^2 & \sigma_A^2 + \sigma_W^2 \rho \\ \sigma_A^2 + \sigma_W^2 \rho & \sigma_A^2 + \sigma_W^2 \end{bmatrix}^{-1} \sigma_A^2 \mathbf{1}$$

$$\mathbf{a} = \frac{\sigma_A^2}{\sigma_W^2} \begin{bmatrix} \frac{\sigma_A^2}{\sigma_W^2} + 1 & \frac{\sigma_A^2}{\sigma_W^2} + \rho \\ \frac{\sigma_A^2}{\sigma_W^2} + \rho & \frac{\sigma_A^2}{\sigma_W^2} + 1 \end{bmatrix}^{-1} \mathbf{1} = \gamma \begin{bmatrix} \gamma+1 & \gamma+\rho \\ \gamma+\rho & \gamma+1 \end{bmatrix}^{-1} \mathbf{1} = \frac{\gamma}{(\gamma+1)^2 - (\gamma+\rho)^2} \begin{bmatrix} \gamma+1 & -(\gamma+\rho) \\ -(\gamma+\rho) & \gamma+1 \end{bmatrix} \mathbf{1}$$

$$a_1 = a_2 = \frac{\gamma}{(\gamma+1)^2 - (\gamma+\rho)^2} (\gamma+1 - (\gamma+\rho)) = \frac{\gamma(1-\rho)}{(\gamma+1)^2 - (\gamma+\rho)^2} = \frac{\gamma}{2\gamma+1+\rho}$$

$$b = (1 - a_1 - a_2) \eta_A = \left(1 - \frac{2\gamma}{2\gamma+1+\rho} \right) \eta_A$$

$$\begin{aligned} \hat{S}_{LMMSE}[n+1] &= \hat{A}_{LMMSE} = \frac{\gamma}{2\gamma+1+\rho} X[n] + \frac{\gamma}{2\gamma+1+\rho} X[n-1] + \left(1 - \frac{2\gamma}{2\gamma+1+\rho} \right) \eta_A \\ &= \alpha \left(\frac{X[n] + X[n-1]}{2} \right) + (1-\alpha) \eta_A = \alpha \bar{X} + (1-\alpha) \eta_A, \end{aligned}$$

$$\text{where } \alpha \triangleq \frac{2\gamma}{2\gamma+1+\rho} \text{ and } \bar{X} \triangleq \frac{X[n] + X[n-1]}{2}$$

(3) MSE of the LMMSE estimator:

$$\begin{aligned} MSE &= E\{\varepsilon^2[n+1]\} = E\{\varepsilon[n+1]S[n+1]\} = E\{(S[n+1] - a_1X[n] - a_2X[n-1] - b)S[n+1]\} \\ &= E\{(S[n+1] - \eta_A - a_1(X[n] - \eta_A) - a_2(X[n-1] - \eta_A))S[n+1]\} \\ &= C_S[0] - a_1C_{SX}[-1] - a_2C_{SX}[-2] = C_S[0] - a_1C_S[1] - a_2C_S[2] \\ &= \sigma_A^2(1 - a_1 - a_2) = \sigma_A^2 \left(1 - \frac{2\gamma}{2\gamma+1+\rho} \right) = \sigma_A^2(1 - \alpha) \end{aligned}$$



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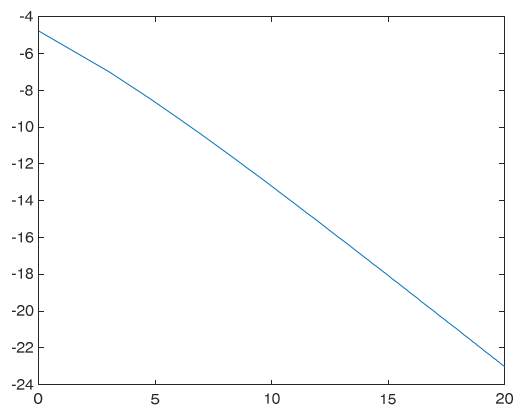
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$$MSE = \sigma_A^2 \left(1 - \frac{2\gamma}{2\gamma+1+\rho} \right) \bigg|_{\sigma_A^2=1, \rho=0} = 1 - \frac{2\gamma}{2\gamma+1} = \frac{1}{2\gamma+1} \leq 10^{-3}$$

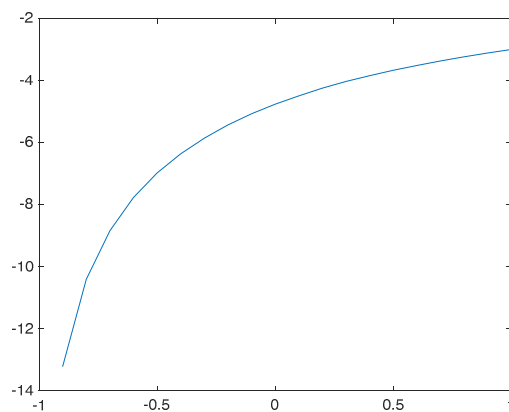
$$\gamma \geq \frac{10^3 - 1}{2} = 499.5$$

(4) LMMSE estimator and MSE as a function of γ and as a function of ρ :

Plot of $MSE_{dB} = 10 \log_{10} MSE = 10 \log_{10} [\sigma_A^2 (1 - \alpha)]$ vs. $\gamma_{dB} = 10 \log_{10} (\gamma)$, for $\sigma_A^2 = 1, \rho = 0$:



Plot of MSE_{dB} vs. γ_{dB} , for $\sigma_A^2 = 1, \gamma = 1$:





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$$\hat{S}_{LMMSE}[n+1] = \alpha \bar{X} + (1-\alpha)\eta_A, \quad MSE = \sigma_A^2(1-\alpha), \quad \alpha = \frac{2\gamma}{2\gamma+1+\rho}$$

$$\sigma_w^2 \rightarrow 0 \Rightarrow \gamma \rightarrow +\infty \Rightarrow \alpha \rightarrow 1 \Rightarrow \hat{S}_{LMMSE}[n+1] \rightarrow \bar{X}, MSE \rightarrow 0$$

$$\sigma_w^2 \rightarrow +\infty \Rightarrow \gamma \rightarrow 0 \Rightarrow \alpha \rightarrow 0 \Rightarrow \hat{S}_{LMMSE}[n+1] \rightarrow \eta_A, MSE \rightarrow \sigma_A^2$$

$$\rho \rightarrow -1 \Rightarrow \alpha \rightarrow 1 \Rightarrow \hat{S}_{LMMSE}[n+1] \rightarrow \bar{X}, MSE \rightarrow 0$$

$$\rho \rightarrow 0 \Rightarrow \alpha \rightarrow \frac{2\gamma}{2\gamma+1} \Rightarrow \hat{S}_{LMMSE}[n+1] \rightarrow \frac{2\gamma}{2\gamma+1} \bar{X} + \left(1 - \frac{2\gamma}{2\gamma+1}\right) \eta_A, MSE \rightarrow \sigma_A^2 \left(1 - \frac{2\gamma}{2\gamma+1}\right) = \frac{\sigma_A^2}{2\gamma+1}$$

$$\rho \rightarrow 1 \Rightarrow \alpha \rightarrow \frac{2\gamma}{2\gamma+2} = \frac{\gamma}{\gamma+1} \Rightarrow \hat{S}_{LMMSE}[n+1] \rightarrow \frac{\gamma}{\gamma+1} \bar{X} + \left(1 - \frac{\gamma}{\gamma+1}\right) \eta_A, MSE \rightarrow \sigma_A^2 \left(1 - \frac{\gamma}{\gamma+1}\right) = \frac{\sigma_A^2}{\gamma+1}$$

Explanation of these behaviors is left to the student.